

CPL Empirical Study: Proposed Revisions Before Submission

Cross-Project Bug Localisation: What Works, What Doesn't, and Why. An Empirical Investigation | A. Eashaan Rao, S. Chimalakonda

Core message. The study seems solid, but the writing has to be revised. It overclaims in places and needs a cleanup pass. The changes below are framing and disclosure, plus a few cheap statistical additions. The core RQ1 result and the controlled multi-instrument protocol are clean and do not need touching.

Content and positioning

1. **Title.** The title is "Cross-Project Bug Localisation: What Works, What Doesn't, and Why. An Empirical Investigation."
2. **Abstract framing.** Reopen on the finding rather than on data-scarcity. Lead with cross-project transfer succeeding or failing as a function of model architecture rather than the source-to-target pair, established through the first controlled multi-instrument evaluation. Keep the data-scarce point, but move it into the introduction as the justification for the WP-small baseline, not as the opening value proposition. This also reduces exposure to the "why not just use an LLM agent" objection.
3. **Overclaims at the claim level.** Two to fix. First, "a model's inference-time adaptation mechanism determines CPL viability more than the source to target pair does" states a law from three architectures. Change "determines" to "appears to be a stronger determinant of." Second, "CP-transfer matches full within-project training" rests on a non-significant $p = 0.19$, which is absence of evidence, not equivalence. Either run a TOST equivalence test on the paired values you already have, or rephrase to "statistically indistinguishable from, though not formally shown equivalent to." Do not leave "matches" on a non-significant result.
4. **Mechanism.** Keep Section 5.2 as descriptive interpretation. Do not name it, box it, or introduce it in the abstract as "we propose." In its closing sentence, change "determines CPL viability" to "appears to govern CPL viability."
5. **Domain-categorisation contradiction (must fix).** Table 2 labels prefect as Systems/Cloud and jupyterlab as Developer Tools/DS, but Table 8 counts all thirty Group A pairs as within-domain DS x DS, and Case 1 calls the prefect to lightning pair within-domain. Pick one taxonomy, recompute Table 8, and re-describe the three cases to match. This moves the TRANSP-CNN 76.7 versus 48.5 percent split that anchors the claim that domain is the primary risk factor for CNN-based CPL, so it has to be internally consistent.
6. **Data leakage (must fix).** BGE-code-v1 was pretrained on a corpus that very likely contains these popular Python repositories, which exposes the frozen-embedding cold-start result. Add a threat paragraph, and if you can, one proxy check, for example whether cold-start MRR correlates with project popularity or star count.
7. **LLM-agent positioning.** Strengthen Section 2.2 with the scoping argument: agents are evaluated on SWE-bench, which selects for well-documented issues with existing tests, while the CPL setting is the complement, that is newly open-sourced, proprietary, privacy-constrained, and sparse-report projects. Optional: one run of Agentless or SWE-agent on a sparse-report target such as xarray at 110 reports, showing underperformance or higher cost, converts the paragraph into a fact.
8. **RQ2 target-LoC finding.** State it as suggestive rather than clean. FAISS recall ranges from 31 to 100 percent, with three projects below 50 percent, which confounds the size result: larger codebase means lower recall, which means less headroom for the reranker to show a gain.
9. **Frozen-backbone confound.** BLAZE freezes its BGE backbone while the other two fine-tune projection layers, which is a confound for the $d = 1.19$ result. State the counterargument once and prominently, that cold-start is 0.262 and not near-optimal, so adapters carry signal, and that the asymmetry is a smoothness property of cosine ranking, rather than splitting it across construct validity and Section 5.2.
10. **Reproducibility (must fix).** Replace the PLACEHOLDER Zenodo DOI with the archived DOI. Add the training hyperparameters that are currently missing: learning rate, optimiser, epochs, and stopping criterion for all three models. Fill in ORCIDs.
11. **Pair count.** The text and tables say 63, but the Figure 6 annotation reads 26/62 and Spearman 0.07, while Table 9 reads 41 percent and 0.076. Make the figure, caption, and table agree.
12. **Factual slips.** scikit-learn is 376K LoC in Table 2 but 316K in Case 3. jupyterlab is called "the smallest project," but docker/compose at 35K is smaller than jupyterlab at 39K. In the Ganin reference, "Ustinova" should be "Ustinova" and "March, M." should be "Marchand, M."

- 13. Ground truth.** Bug-report-to-file mappings taken from commit SHAs carry the tangled-commit problem, where a fixing commit also touches unrelated files. The filter on multi-file patches over ten files helps. Add one sentence naming the residual noise as a construct threat.

Writing cleanup

- 14. Em dashes.** Remove all of them. They appear in two roles. For a clause set off mid-sentence, use commas, or parentheses if the aside is secondary. For a dash introducing a consequence or restatement, use a comma, a colon, or split into two sentences. Use a colon when a list or an unpacking follows, for example "a consistent risk spectrum: negative transfer rates of 3.2, 27.0, and 31.7 percent." Do a global find on the em dash character and resolve each by hand, since the right replacement depends on the role. Keep en dashes in numeric ranges such as 35K to 572K.
- 15. "Significant."** Keep it only where a statistical test sits next to it. Where it means "large" or "important" with no test behind it, replace it with the concrete claim or a neutral word. A reviewer who sees "significant" without a test starts doubting the ones that do have tests.
- 16. Unearned qualifiers.** Remove "directly" wherever it is doing rhetorical rather than literal work, for example "the contributions of this work directly address" becomes "the contributions of this work address." Sweep the same for "substantially," "dramatically," "markedly," "clearly," "consistently," and "robustly." If deleting the word does not change what is true, delete it. "robustly" in particular should go, so "scales robustly across the full range" becomes "remains positive across all size groups."
- 17. Novelty phrasing.** "the first," "no existing study," and "to our knowledge" recur many times. Keep them for the two claims that matter, the multi-instrument protocol and the WP-small baseline, and soften the repeats to "we are not aware of."

The rule for both sweeps in one line: remove every em dash, and keep "significant" only where a p-value sits beside it.

Figure

- 18. Add an end-to-end study-design figure as Figure 1** at the top of Section 3. The paper has result figures but no single overview of the pipeline, so a reader has to assemble the design from prose. Show, left to right: corpus construction (five benchmarks, funnel to the 13 Python projects with the three diversity criteria), pair selection (13 projects into 63 directional pairs, Groups A, B, C), the shared infrastructure (BGE-code-v1 embedding into the FAISS top-300 stage, drawn once as the common backbone), the three models as parallel branches off the shared candidate pool with their ranking mechanisms labelled, the four training scenarios as a band across the models with the test set held constant, and the evaluation stage (five metrics, then the three RQ analyses). The point the figure must make visible is the controlled design: one embedding backbone and one retrieval stage feeding three architectures under one set of scenarios and one baseline, so that everything is held constant except the ranking mechanism. Keep it a black-and-white process figure, boxes and flow, readable at column width, not a results figure.